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Calculation of differential seismograms using analytic partial derivatives – I: teleseismic receiver functions

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SUMMARY

Teleseismic receiver function inversion for crustal and upper-mantle velocity structure is intrinsically nonlinear and is often solved by iterative linearization methods. This requires calculation of partial derivatives of receiver functions with respect to model parameters which has been done by finite differences so far. Here we derived analytic partial derivatives of receiver functions with respect to different model parameters in multilayered velocity model using the Haskell propagation matrix formalism. We also implemented a speed-up algorithm to compute differential receiver functions by storing intermediate products of Haskell matrices so that the total computation time is linearly proportional to the number of layers in the model. Numerical tests show that our implementation is correct, numerically stable and efficient. Using more accurate analytic partial derivatives in receiver function inversions helps iterative inversion to converge more rapidly than using finite-differences partial derivatives. We also found that the optimal perturbation size in computing finite-differences partial derivatives of receiver functions is between 0.01 per cent and 0.1 per cent. Our results should be applicable to a range of geophysical inverse problems involving receiver functions.

Key words: Waveform inversion; Body waves; Theoretical seismology; Wave propagation.

1 INTRODUCTION

Teleseismic receiver functions (RF) have been widely used to determine crustal and upper mantle structure of the Earth. With removal of earthquake's source time function as well as propagation effects on the source side and in the lower mantle, RFs contain rich information about the seismic velocity structure on the receiver side (e.g. Vinnik 1977; Langston 1979). Early use of RFs was focused on inversion for local 1-D shear wave velocity structure beneath the station (e.g. Owens *et al.* 1984; Zhu *et al.* 1993). To deal with the non-uniqueness of inversion using RFs alone (Ammon *et al.* 1990), the state-of-the-art methodology is to combine other data sets with RFs for joint inversion, including surface wave phase and group velocities (e.g Özalaybey *et al.* 1997; Du & Foulger 1999; Julia *et al.* 2000), Rayleigh wave ellipticity (e.g. Chong *et al.* 2016; Kang *et al.* 2016), body-wave traveltimes (e.g. Vinnik *et al.* 2006), gravity data (e.g. Chai *et al.* 2015) and magnetotelluric data (e.g. Moorkamp *et al.* 2007).

While inversion problems involving RFs are intrinsically nonlinear and are sometimes solved using stochastic or probabilistic inversion methods (e.g. Shibutani *et al.* 1996; Zhao *et al.* 1996; Sambridge 1999; Shen *et al.* 2013), more often iterative linearized inversion approaches are used for computational efficiency (e.g. Owens *et al.* 1984; Julia *et al.* 2000, 2009; Chai *et al.* 2015). Linearization requires computation of partial derivatives of RFs with respect to the model parameters and the accuracy of calculation is critical in success of the inversion. So far computation of RF partial derivatives has been done numerically using the finite differences (FD) method (e.g. Owens *et al.* 1984; Randall 1989; Herrmann 2013), that is, by finitely perturbing model parameters and computing differences of RFs. Theoretically, the perturbation should be infinitely small to obtain accurate partial derivatives. This is, however, impractical due to fixed number of significant digits in computer's representation of real numbers. The perturbation magnitude is often chosen on an *ad hoc* basis and assessment of impact on accuracy is difficult.

In this study, we derive analytic partial derivatives of RFs with respect to model parameters. We first express teleseismic RFs due to a plane wave impinging on a multilayered medium using the Thomson–Haskell propagator matrix method (Haskell 1953, 1963; Zhu & Rivera 2002). We then derive analytic partial derivatives of the propagator matrix in each layer with respect to the seismic velocities and density in the layer. When computing RF partial derivatives in a multilayered model, we follow the algorithm of Randall (1989) to store intermediate

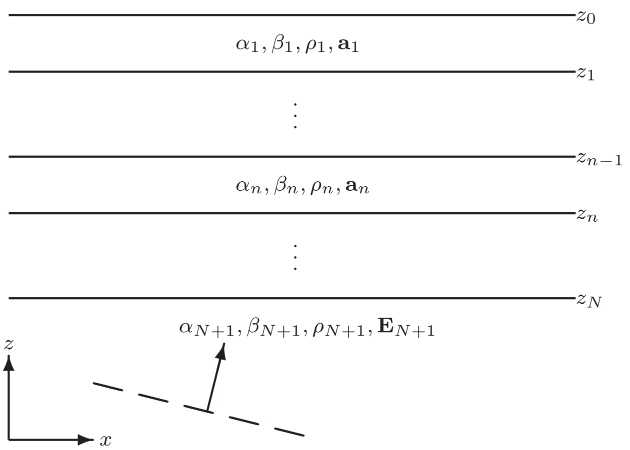

Figure 1. A plane wave impinging on a multilayered velocity model of N layers over a half-space.

results of product of propagator matrices to reduce computational time. Numerical tests and comparison with FD results show that our implementation is correct and efficient.

2 THEORY

2.1 Theoretical RFs of multilayered model

For a multilayered model shown in Fig. 1 and an incidence P or SV plane wave in the bottom half-space, we choose a Cartesian coordinate system with the z-axis in the vertical direction and the x-axis in the horizontal wave propagation direction. The P–SV displacement wavefield generated by the incident plane wave can be expressed as

$$\mathbf{u}(x, z, t) = \int (A_x(z, \omega)\hat{\mathbf{e}}_x + A_z(z, \omega)\hat{\mathbf{e}}_z)\, \mathrm{e}^{\mathrm{i}(\omega t - kx)} \mathrm{d}\omega, \tag{1}$$

where k is the horizontal wavenumber of the plane wave, A_z and A_x are displacement components in the vertical and radial directions, respectively, in the frequency domain. Similar expression can be written for the traction vector,

$$\sigma(x, z, t) = k \int (T_x(z, \omega)\hat{\mathbf{e}}_x + T_z(z, \omega)\hat{\mathbf{e}}_z)\, \mathrm{e}^{\mathrm{i}(\omega t - kx)} \mathrm{d}\omega. \tag{2}$$

Substituting $\mathbf{u}$ and σ in the second-order differential equation of motion for the P–SV system and the stress-strain elasticity relationship results in a set of first-order ordinary differential equations:

$$\frac{\mathrm{d}\mathbf{b}(z)}{\mathrm{d}z} = \mathbf{M}\mathbf{b}(z), \tag{3}$$

where

$$\mathbf{b}(z) = [\mathrm{i}A_x, A_z, T_z, \mathrm{i}T_x]^T, \tag{4}$$

$$\mathbf{M} = k \begin{bmatrix} 0 & -1 & 0 & \dfrac{1}{\mu} \\ 1 - 2\xi & 0 & \dfrac{\xi}{\mu} & 0 \\ 0 & -\rho\left(\dfrac{\omega}{k}\right)^2 & 0 & 1 \\ 4\mu\xi_1 - \rho\left(\dfrac{\omega}{k}\right)^2 & 0 & 2\xi - 1 & 0 \end{bmatrix}, \tag{5}$$

ρ is the density; $\xi = \mu/(\lambda + 2\mu)$; $\xi_1 = 1 - \xi$; λ, μ are the Lame constants. $\mathbf{b}$ is called the displacement-stress vector. Here, the imaginary unit i is multiplied to the horizontal components A_x and T_x in $\mathbf{b}$ so that the z-independent matrix $\mathbf{M}$ is the same as in the cylindrical coordinate system (Herrmann 2013). The general solution to Eq. (3) in a uniform layer can be written as (e.g. Zhu & Rivera 2002)

$$\mathbf{b}(z) = \mathbf{E}\Lambda(z)\mathbf{w}, \tag{6}$$

where

$$\mathbf{E} = \begin{bmatrix} -1 & -\dfrac{\nu_\beta}{k} & 1 & \dfrac{\nu_\beta}{k} \\ \dfrac{\nu_\alpha}{k} & 1 & \dfrac{\nu_\alpha}{k} & 1 \\ -2\,\mu\,\gamma_1 & -2\,\mu\,\dfrac{\nu_\beta}{k} & 2\,\mu\,\gamma_1 & 2\,\mu\,\dfrac{\nu_\beta}{k} \\ 2\,\mu\,\dfrac{\nu_\alpha}{k} & 2\,\mu\,\gamma_1 & 2\,\mu\,\dfrac{\nu_\alpha}{k} & 2\,\mu\,\gamma_1 \end{bmatrix}, \tag{7}$$

$$\Lambda(z) = \begin{bmatrix} \mathrm{e}^{-\nu_\alpha z} & 0 & 0 & 0 \\ 0 & \mathrm{e}^{-\nu_\beta z} & 0 & 0 \\ 0 & 0 & \mathrm{e}^{\nu_\alpha z} & 0 \\ 0 & 0 & 0 & \mathrm{e}^{\nu_\beta z} \end{bmatrix}, \tag{8}$$

$$\mathbf{w} = \left[w_P^U, w_{SV}^U, w_P^D, w_{SV}^D\right]^T, \tag{9}$$

and

$$\nu_\alpha = \sqrt{k^2 - k_\alpha^2}, \quad k_\alpha = \frac{\omega}{\alpha}, \tag{10}$$

$$\nu_\beta = \sqrt{k^2 - k_\beta^2}, \quad k_\beta = \frac{\omega}{\beta}, \tag{11}$$

$$\gamma = 2\frac{k^2}{k_\beta^2}, \quad \gamma_1 = 1 - \frac{1}{\gamma}. \tag{12}$$

α, β are the compressional and shear velocities of the layer. $w_{P,SV}^{U,D}$,s are the potential coefficients of up and down-going P and SV waves in the layer, to be determined by the boundary conditions. The displacement-stress vectors at the top and bottom of the layer are related by,

$$\mathbf{b}(z_n) = \mathbf{a}_n \mathbf{b}(z_{n-1}), \tag{13}$$

where

$$\mathbf{a}_n = \mathbf{E}_n \Lambda(z_n - z_{n-1})\mathbf{E}_n^{-1}, \tag{14}$$

is called the Thomson–Haskell propagator matrix (Haskell 1953, 1963). Its components can be found in Zhu & Rivera (2002).

The surface displacement-stress vector $\mathbf{b}_0$ can be obtained by the traction-free surface boundary condition and displacement-stress continuity across all layer interfaces,

$$\mathbf{a}_N \cdots \mathbf{a}_1 \mathbf{b}_0 = \mathbf{E}_{N+1}\Lambda(z_N)\mathbf{w}_N. \tag{15}$$

So the displacement components at the surface

$$\begin{bmatrix} \mathrm{i}A_x \\ A_z \end{bmatrix} = \frac{1}{R|_{12}^{12}} \begin{bmatrix} R_{22} & -R_{12} \\ -R_{21} & R_{11} \end{bmatrix} \begin{bmatrix} w_P^U\ \mathrm{e}^{-\nu_\alpha z_N} \\ w_{SV}^U \mathrm{e}^{-\nu_\beta z_N} \end{bmatrix}, \tag{16}$$

where

$$\mathbf{R} = \mathbf{E}_{N+1}^{-1}\mathbf{a}_N \cdots \mathbf{a}_1, \tag{17}$$

and

$$R|_{12}^{12} = R_{11}R_{22} - R_{12}R_{21}, \tag{18}$$

is the Rayleigh denominator.

For a P incident plane wave in the bottom half-space, $w_{SV}^U = 0$. So the P RF in the frequency domain is

$$F^P(\omega) = \frac{A_x}{A_z} = \mathrm{i}\frac{R_{22}}{R_{21}}. \tag{19}$$

For an SV incident plane wave, $w_P^U = 0$ and the S RF is

$$F^S(\omega) = \frac{A_z}{A_x} = -\mathrm{i}\frac{R_{11}}{R_{12}}. \tag{20}$$

2.2 Analytic partial derivatives of RFs

Let m represent an independent model parameter, for example, α, β or ρ. Using the above results, the partial derivative of RF with respect m can be obtained using the quotient rule,

$$F^P{}_{,m} = \mathrm{i}\frac{R_{22,m}R_{21} - R_{21,m}R_{22}}{R_{21}^2}, \tag{21}$$

$$F^S{}_{,m} = -\mathrm{i}\frac{R_{11,m}R_{12} - R_{12,m}R_{11}}{R_{12}^2}, \tag{22}$$

where

$$\mathbf{R}_{,m} = \begin{cases} \mathbf{E}_{N+1}^{-1}\mathbf{a}_N \cdots \mathbf{a}_{n,m} \cdots \mathbf{a}_1, & \text{m in the n-th layer,} \\ \mathbf{E}_{N+1,m}^{-1}\mathbf{a}_N \cdots \mathbf{a}_1, & \text{m in the half-space.} \end{cases} \tag{23}$$

Components of $\mathbf{E}$ and $\mathbf{a}$ were given in Zhu & Rivera (2002). Their partial derivatives with respect to different model parameters in seismology are given in Appendix A.

3 IMPLEMENTATION AND TESTS

Eqs (21)–(23) show that computation of RF partial derivative with respect to one layer parameter requires calculation of the partial derivative of the layer Haskell matrix and its product with Haskell matrices of all other layers. So the total computation time of all RF partial derivatives for an N-layer model is proportional to N^2, that is, the computation complexity is $O(N^2)$. Randall (1989) proposed an optimization algorithm to reduce the complexity to $O(N)$ by storing the intermediate products of Haskell matrices so that they can be repeatedly used. We implemented Randall (1989)'s optimization in our computation of $\mathbf{R}_{,m}$. It first computes the Haskell matrix of each layer in a loop from the bottom to top of the model. Products of the Haskell matrices from the bottom layer to all other layers are stored. It then computes the partial derivatives of $\mathbf{R}$ with respect to each layer in a second loop from the top to the bottom, using Eq. (23) with the stored Haskell matrices and their products. Fig. 2 shows computation times of brute-force FD method and our implementation. The computation time of our implementation is linearly proportional to the number of layers while the computation time of the brute-force FD method is $O(N^2)$. When the number of layers of the model is 30, the optimization can save approximately 90 per cent of computation time.

To check the correctness of our implementation of analytic partial derivatives, we compared differential P and S RFs from using the FD method and analytic partial derivatives. We used a 1-D southern California velocity model by Hadley & Kanamori (1977) (Table 1). A small perturbation of 0.1 per cent of model parameters was used in the FD computation. All RFs were low-pass filtered at around 1.0 Hz using a Gaussian parameter of 3. As shown in Fig. 3, waveforms of FD and analytic partial derivative RFs agree well with each other, confirming that our analytic derivatives are correct.

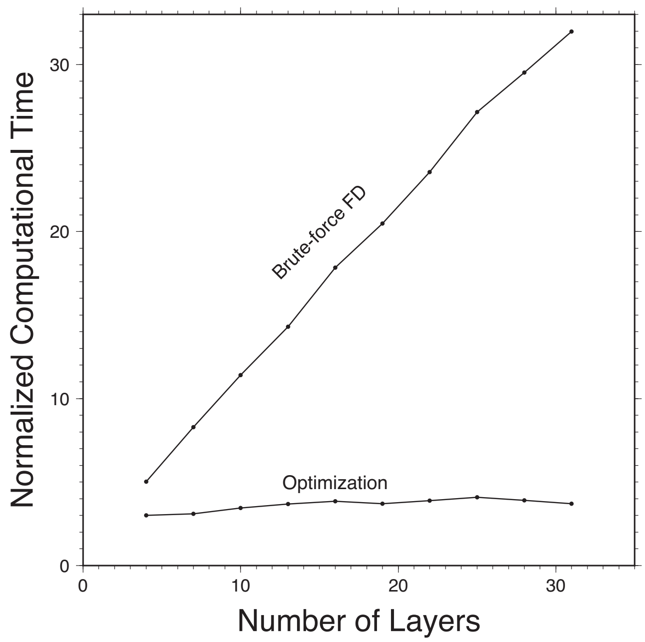

Figure 2. Comparison of total computation times of differential RFs using brute-force FD and analytic partial derivatives with optimization (see the text). The computational times are normalized by the computation time of receiver function of the corresponding velocity model.

Table 1. The Hadley–Kanamori velocity model.

Th. (km)	β (km s^{-1})	α (km s^{-1})	ρ (g cm^{-3})	Q_β	Q_α
5.5	3.18	5.50	2.53	600	1200
10.5	3.64	6.30	2.79	600	1200
16.0	3.87	6.70	2.91	600	1200
	4.50	7.80	3.27	900	1800

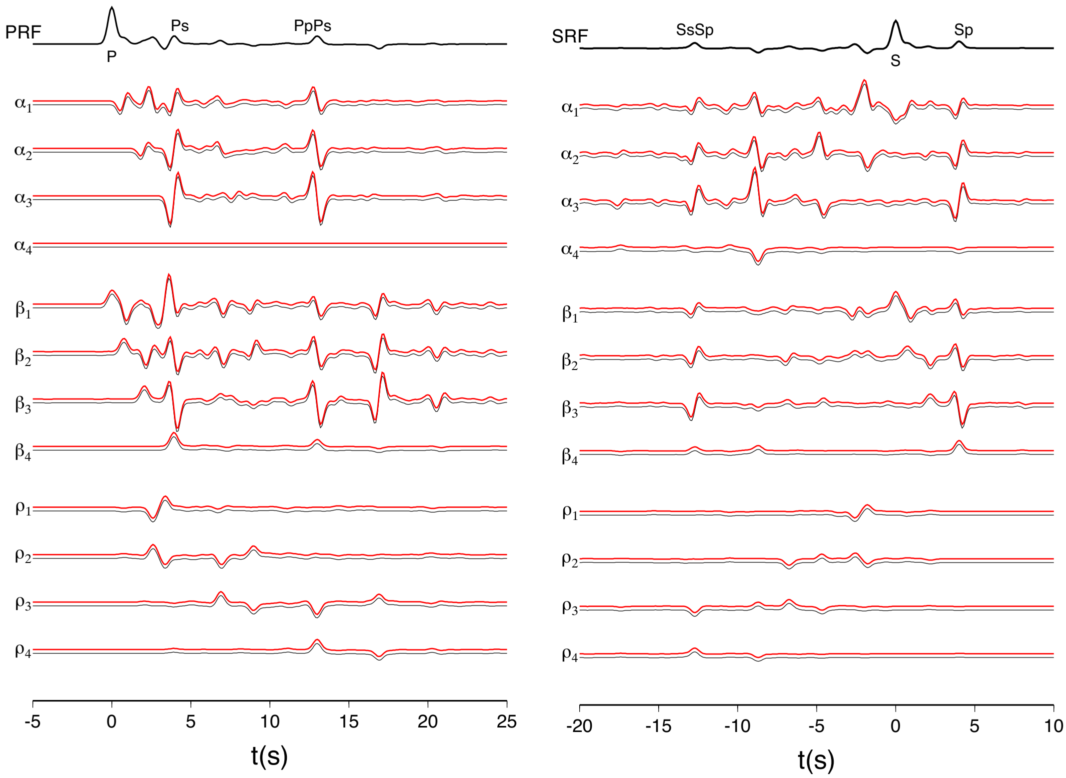

Figure 3. Comparison of differential P and S RFs using the FD method (black) and analytic partial derivatives (red) with respect to velocities and density of each layer. A ray parameter 0.07 s km^{-1} was used for PRF and 0.08 s km^{-1} for SRF. P-to-S and S-to-P converted phases at the Moho and its crustal multiples are labelled. A finite perturbation of 0.1 per cent was used for the FD method. Amplitudes of differential RFs with respect to α are amplified three times for visualization. Note that the PRF is independent of α in the half-space.

We tested using FD and analytic derivative RFs in RF inversion with the iterative damped least-squares method. Four RFs with ray parameters between 0.04 and 0.08 s/km were generated as data using the Hadley-Kanamori velocity model. They were low-pass filtered at ∼1 Hz using a Gaussian parameter of 3. Only shear-wave velocities were determined with a fixed α/β ratio of 1.73 and an empirical relationship $\rho = 0.77 + 0.32\alpha$ between the density and α (Berteussen 1977; Owens *et al.* 1984). We started with a simple one-layer-over-half-space crustal model. In both cases, the damped least-squares inversions fully recovered the true velocity model after certain number of iterations. But the inversion using analytic derivatives converges more rapidly than the inversion using FD derivatives (Fig. 4). The predicted RF waveforms fit the 'observations' very well.

We also repeated the above test using real RFs of station SLM, a permanent broadband seismic station located in St. Louis, Missouri, USA (http://www.eas.slu.edu/eqc/eqc_netinfo/SLM/slm.html). Five RFs with ray parameters between 0.053 and 0.079 s/km from teleseismic earthquakes in the back-azimuths of 160°–166° were used. The inversion started with a 1-D central US velocity model (CUS; Wang & Herrmann 1980). To compute FD partial derivatives, we tested two perturbation sizes, 5 per cent and 10 per cent. Iterative inversions using the analytic and 5 per cent-FD partial derivatives reduced waveform misfits to the same level and obtained the same final model, but the former converged fast (Fig. 5). The final model is similar to CUS above 40 km depth but has a deeper Moho (∼56 km) beneath a high-velocity lower crust. The crustal thickness agrees with results of recent RF study using USArray stations in the area (McGlannan & Gilbert 2016). Iterative inversion using less accurate FD partial derivatives of 10 per cent perturbation ended in a local minimum with about 1 per cent higher RMS misfit (Fig. 5).

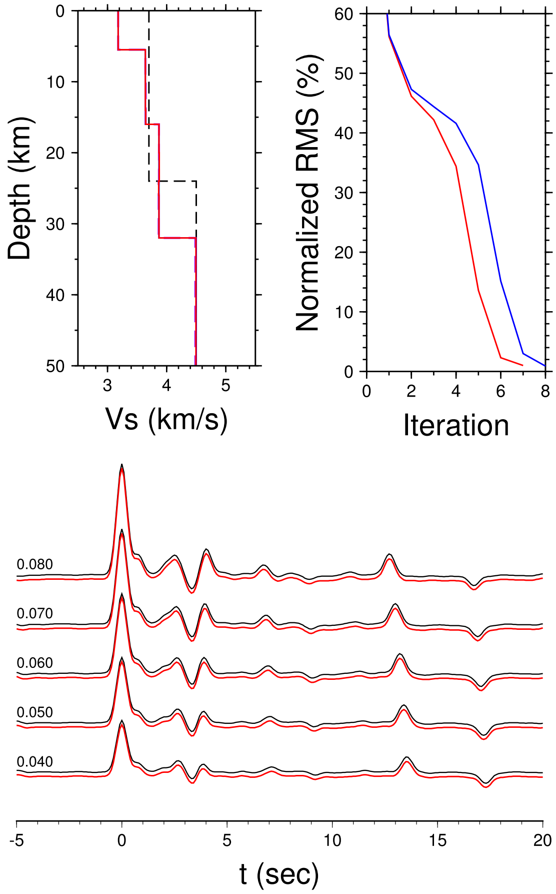

Figure 4. RF inversion results of damped least squares with analytic derivatives and FD derivatives. Top-left: the initial model (dashed line), obtained final model (red), and true model (black). Top-right: decrease of waveform misfits with iteration in inversions using analytic partial derivatives (red) and FD (blue). A finite perturbation of 5 per cent was used in the FD calculation. Bottom: waveform fits of 'observed' RFs (black) and predictions of the final model (red). Ray parameters of RFs are shown in s km^{-1} to the left of the traces.

3.1 Discussion and conclusions

With analytic partial derivatives of RFs available, we can now determine the optimal finite perturbation of model parameters for computing differential RFs using the FD method. We tried a range of perturbation sizes from 0.0001 per cent to 5 per cent for α, β and ρ and calculated FD differential RFs of the Hadley–Kanamori model. Root mean squares (RMS) of waveform differences between the FD and analytic differential RFs were then calculated. Their variations with the perturbation size show that the optimal perturbation is between 0.01 per cent and 0.1 per cent for velocities and between 0.01 per cent to 1 per cent for density (Fig. 6). Too small perturbations lead to large computation error due to loss of significant digits while large perturbations decrease FD accuracy.

Our analytic RF partial derivatives were derived using the Haskell propagator matrix formalism. Since the Haskell matrix contains exponential functions like $e^{\pm\nu d}$, multiplication of propagator matrices of several layers may cause severe loss of significant digits and large numerical errors. One solution is to use the so-called compound matrix instead of the Haskell matrix itself (e.g. Dunkin 1965; Watson 1970; Wang & Herrmann 1980). Another solution is to partition the propagator matrix into generalized reflection-transmission coefficient matrices (Kennett 1983) which has been used in most theoretical RF calculation (e.g. Owens *et al.* 1984; Randall 1989). However, concern of numerical instability in computing the Haskell matrices is not needed for teleseismic incident waves where the ray parameters are so small that the vertical wavenumbers $\nu_{\alpha,\beta}$ are purely imaginary. We tested an upper-mantle model of Parametric Earth Models (PEM-C) (Dziewonski *et al.* 1975) that consists of 59 layers to a depth of 250 km (Fig. 7). The results confirm that the Haskell propagator matrix formalism is numerically stable for both *P* and *S* RFs of different incident angles down to upper mantle depth.

In summary, we derived analytic partial derivatives of RFs with respect to different model parameters in multilayered velocity model using the Haskell propagation matrix formalism. We also implemented a speed-up algorithm to compute differential RFs by storing intermediate products of Haskell matrices so that the total computation time is linearly proportional to the number of layers in the model. Numerical tests show our implementation is correct, numerically stable, and efficient. Using more accurate analytic partial derivatives in RF inversions helps iterative inversion to converge more rapidly than using to FD partial derivatives. We also found that the optimal perturbation size in computing FD partial derivative of RFs is between 0.01 per cent and 0.1 per cent. Our results should be applicable to a range of geophysical inverse problems involving RFs.

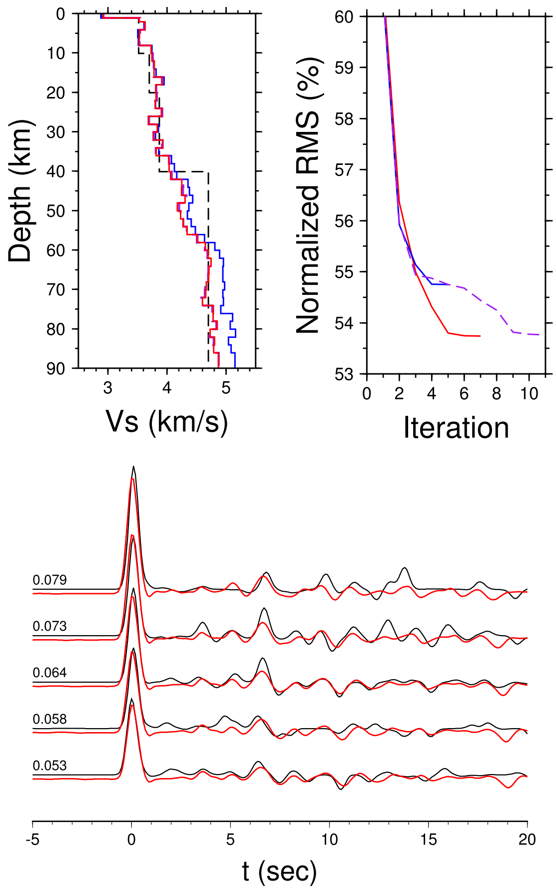

Figure 5. Inversion results of damped least-squares using RFs of station SLM. Top-left: the initial model (dashed line), and obtained models using analytic derivatives (red) and FD derivatives of perturbation steps of 5 per cent (red) and 10 per cent (blue). Top-right: decrease of waveform misfits with iteration in inversions using analytic partial derivatives (red) as well as FD of 5 per cent (purple) and 10 per cent perturbation (blue). Bottom: waveform fits of observed RFs (black) and predictions of the best final model (red). Ray parameters of RFs are shown in s km^{-1} to the left of the traces.

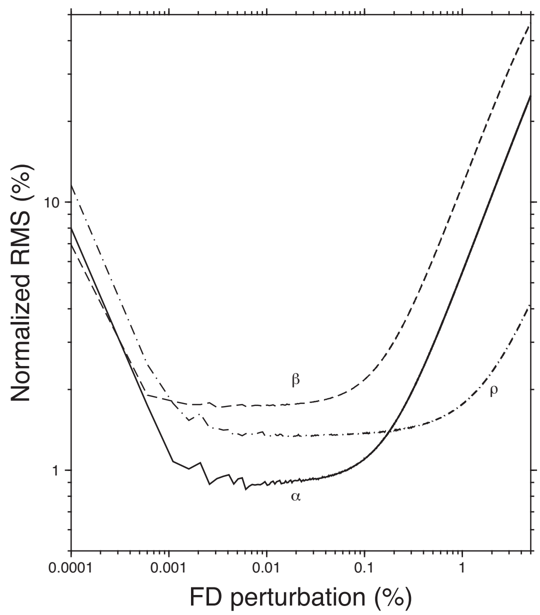

Figure 6. Root mean squares (RMS) of waveform differences between FD and analytic differential RFs. The RMSs are normalized using the RMS of receiver functions. A ray parameter of 0.07 s km^{-1} was used.

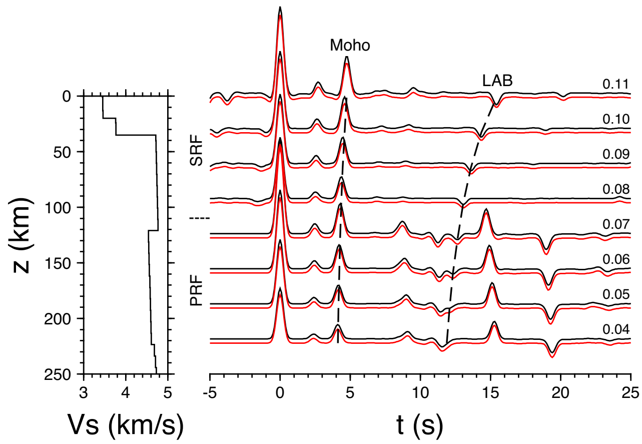

Figure 7. Comparison of *P* and *S* RFs using the Haskell propagator matrix formalism (red) and Kennett's generalized reflection-transmission coefficient matrix formalism (black). The upper-mantle PEM-C velocity model used is shown on the left, and the ray parameters are from 0.04 to 0.07 s km^{-1} for PRF and from 0.08 to 0.11 s km^{-1} for SRF (labelled next to the traces). *P*-to-*S* or *S*-to-*P* converted phases from the Moho and lithosphere-asthenosphere boundary (LAB) are labelled.

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APPENDIX A: PARTIAL DERIVATIVES OF THE HASKELL MATRIX

Components of Haskell matrix **a** and $\mathbf{E}^{-1}$ were given in Zhu & Rivera (2002). For commonly-used independent layer parameters of α, β, and ρ, the non-zero partial derivatives of Haskell matrix components are:

$$\frac{\partial a_{11}}{\partial \alpha} = \frac{1}{\alpha} kdY_\alpha \gamma_2, \tag{A1}$$

$$\frac{\partial a_{12}}{\partial \alpha} = \frac{1}{\alpha}\frac{k^2}{\nu_\alpha^2}(kdC_\alpha - Y_\alpha)\gamma_1\gamma_2, \tag{A2}$$

$$\frac{\partial a_{13}}{\partial \alpha} = -\frac{1}{2\mu\alpha} kdY_\alpha \gamma_2, \tag{A3}$$

$$\frac{\partial a_{14}}{\partial \alpha} = -\frac{1}{2\mu\alpha}\frac{k^2}{\nu_\alpha^2}(kdC_\alpha - Y_\alpha)\gamma_2, \tag{A4}$$

$$\frac{\partial a_{21}}{\partial \alpha} = -\frac{1}{\alpha}(kdC_\alpha + Y_\alpha)\gamma_2, \tag{A5}$$

$$\frac{\partial a_{22}}{\partial \alpha} = -\frac{1}{\alpha} kdY_\alpha \gamma_1\gamma_2, \tag{A6}$$

$$\frac{\partial a_{23}}{\partial \alpha} = \frac{1}{2\mu\alpha}(kdC_\alpha + Y_\alpha)\gamma_2, \tag{A7}$$

$$\frac{\partial a_{24}}{\partial \alpha} = \frac{1}{2\mu\alpha} kdY_\alpha \gamma_2, \tag{A8}$$

$$\frac{\partial a_{31}}{\partial \alpha} = \frac{2\mu}{\alpha} kdY_\alpha \gamma_1\gamma_2, \tag{A9}$$

$$\frac{\partial a_{32}}{\partial \alpha} = \frac{2\mu}{\alpha}\frac{k^2}{\nu_\alpha^2}(kdC_\alpha - Y_\alpha)\gamma_1^2\gamma_2, \tag{A10}$$

$$\frac{\partial a_{33}}{\partial \alpha} = -\frac{1}{\alpha} kdY_\alpha \gamma_1\gamma_2, \tag{A11}$$

$$\frac{\partial a_{34}}{\partial \alpha} = -\frac{1}{\alpha}\frac{k^2}{\nu_\alpha^2}(kdC_\alpha - Y_\alpha)\gamma_1\gamma_2, \tag{A12}$$

$$\frac{\partial a_{41}}{\partial \alpha} = -\frac{2\mu}{\alpha}(kdC_\alpha + Y_\alpha)\gamma_2, \tag{A13}$$

$$\frac{\partial a_{42}}{\partial \alpha} = -\frac{2\mu}{\alpha} kdY_\alpha \gamma_1\gamma_2, \tag{A14}$$

$$\frac{\partial a_{43}}{\partial \alpha} = \frac{1}{\alpha}(kdC_\alpha + Y_\alpha)\gamma_2, \tag{A15}$$

$$\frac{\partial a_{44}}{\partial \alpha} = \frac{1}{\alpha} kdY_\alpha \gamma_2, \tag{A16}$$

$$\frac{\partial a_{11}}{\partial \beta} = \frac{2}{\beta}[\gamma(C_\alpha - C_\beta) - \gamma_1 kdY_\beta], \tag{A17}$$

$$\frac{\partial a_{12}}{\partial \beta} = \frac{2}{\beta}[\gamma(Y_\alpha - X_\beta) - (kdC_\beta + Y_\beta)], \tag{A18}$$

$$\frac{\partial a_{13}}{\partial \beta} = \frac{1}{\mu\beta} kdY_\beta, \tag{A19}$$

$$\frac{\partial a_{14}}{\partial \beta} = \frac{1}{\mu\beta}(kdC_\beta + Y_\beta), \tag{A20}$$

$$\frac{\partial a_{21}}{\partial \beta} = \frac{2\gamma}{\beta}[(Y_\beta - X_\alpha) + \gamma_1\gamma_3(kdC_\beta - Y_\beta)], \tag{A21}$$

$$\frac{\partial a_{22}}{\partial \beta} = \frac{2}{\beta}[\gamma(C_\beta - C_\alpha) + kdY_\beta], \tag{A22}$$

$$\frac{\partial a_{23}}{\partial \beta} = \frac{1}{\mu\beta}(kdC_\beta - Y_\beta)\gamma\gamma_3, \tag{A23}$$

$$\frac{\partial a_{24}}{\partial \beta} = -\frac{\partial a_{13}}{\partial \beta}, \tag{A24}$$

$$\frac{\partial a_{31}}{\partial \beta} = \frac{4\mu}{\beta}[(2\gamma - 1)(C_\alpha - C_\beta) - \gamma_1 kdY_\beta], \tag{A25}$$

$$\frac{\partial a_{32}}{\partial \beta} = \frac{4\mu}{\beta}[2\gamma(\gamma_1 Y_\alpha - X_\beta) - (kdC_\beta + Y_\beta)], \tag{A26}$$

$$\frac{\partial a_{33}}{\partial \beta} = \frac{\partial a_{22}}{\partial \beta}, \tag{A27}$$

$$\frac{\partial a_{34}}{\partial \beta} = -\frac{\partial a_{12}}{\partial \beta}, \tag{A28}$$

$$\frac{\partial a_{41}}{\partial \beta} = \frac{4\mu\gamma}{\beta}\left[2\gamma_1 Y_\beta - 2X_\alpha + \gamma_1^2\gamma_3(kdC_\beta - Y_\beta)\right], \tag{A29}$$

$$\frac{\partial a_{42}}{\partial \beta} = -\frac{\partial a_{31}}{\partial \beta}, \tag{A30}$$

$$\frac{\partial a_{43}}{\partial \beta} = -\frac{\partial a_{21}}{\partial \beta}, \tag{A31}$$

$$\frac{\partial a_{44}}{\partial \beta} = \frac{\partial a_{11}}{\partial \beta}, \tag{A32}$$

$$\frac{\partial a_{13}}{\partial \rho} = -\frac{\gamma}{2\rho\mu}(-C_\alpha + C_\beta), \tag{A33}$$

$$\frac{\partial a_{14}}{\partial \rho} = -\frac{\gamma}{2\rho\mu}(-Y_\alpha + X_\beta), \tag{A34}$$

$$\frac{\partial a_{23}}{\partial \rho} = -\frac{\gamma}{2\rho\mu}(X_\alpha - Y_\beta), \tag{A35}$$

$$\frac{\partial a_{24}}{\partial \rho} = -\frac{\gamma}{2\rho\mu}(C_\alpha - C_\beta), \tag{A36}$$

$$\frac{\partial a_{31}}{\partial \rho} = \frac{2\mu\gamma\gamma_1}{\rho}(C_\alpha - C_\beta), \tag{A37}$$

$$\frac{\partial a_{32}}{\partial \rho} = \frac{2\mu\gamma}{\rho}\left(\gamma_1^2 Y_\alpha - X_\beta\right), \tag{A38}$$

$$\frac{\partial a_{41}}{\partial \rho} = \frac{2\mu\gamma}{\rho}\left(-X_\alpha + \gamma_1^2 Y_\beta\right), \tag{A39}$$

$$\frac{\partial a_{42}}{\partial \rho} = \frac{2\mu\gamma\gamma_1}{\rho}(-C_\alpha + C_\beta), \tag{A40}$$

where

$$\gamma_2 = \gamma\frac{k_\alpha^2}{k^2}, \quad \gamma_3 = \frac{1}{\gamma - 2}, \tag{A41}$$

$$C_\alpha = \cosh(\nu_\alpha d), \quad C_\beta = \cosh(\nu_\beta d), \tag{A42}$$

$$X_\alpha = \frac{\nu_\alpha}{k}\sinh(\nu_\alpha d), \quad X_\beta = \frac{\nu_\beta}{k}\sinh(\nu_\beta d), \tag{A43}$$

$$Y_\alpha = \frac{k}{\nu_\alpha}\sinh(\nu_\alpha d), \quad Y_\beta = \frac{k}{\nu_\beta}\sinh(\nu_\beta d). \tag{A44}$$

The partial derivatives of $\mathbf{E}^{-1}$ are:

$$\frac{\partial \mathbf{E}^{-1}}{\partial \alpha} = \frac{1}{\alpha}\frac{k_\alpha^2 k^3}{k_\beta^2 \nu_\alpha^3}\begin{bmatrix} 0 & \gamma_1 & 0 & -\frac{1}{2\mu} \\ 0 & 0 & 0 & 0 \\ 0 & \gamma_1 & 0 & -\frac{1}{2\mu} \\ 0 & 0 & 0 & 0 \end{bmatrix}, \tag{A45}$$

$$\frac{\partial \mathbf{E}^{-1}}{\partial \beta} = \frac{\gamma}{\beta}\begin{bmatrix} -1 & -\frac{k}{\nu_\alpha} & 0 & 0 \\ \frac{k}{\nu_\beta}(1-\gamma_1\gamma_3) & 1 & \frac{k\gamma_3}{2\mu\nu_\beta} & 0 \\ 1 & -\frac{k}{\nu_\alpha} & 0 & 0 \\ -\frac{k}{\nu_\beta}(1-\gamma_1\gamma_3) & 1 & -\frac{k\gamma_3}{2\mu\nu_\beta} & 0 \end{bmatrix}, \tag{A46}$$

$$\frac{\partial \mathbf{E}^{-1}}{\partial \rho} = \frac{\gamma}{4\rho\mu}\begin{bmatrix} 0 & 0 & -1 & -\frac{k}{\nu_\alpha} \\ 0 & 0 & \frac{k}{\nu_\beta} & 1 \\ 0 & 0 & 1 & -\frac{k}{\nu_\alpha} \\ 0 & 0 & -\frac{k}{\nu_\beta} & 1 \end{bmatrix}. \tag{A47}$$

If $\kappa = \alpha/\beta$ is used as an independent parameter instead of α, the partial derivatives with respect to κ and β can be obtained using the following relationships,

$$\left.\frac{\partial F}{\partial \beta}\right|_\kappa = \kappa\frac{\partial F}{\partial \alpha} + \frac{\partial F}{\partial \beta}, \tag{A48}$$

$$\left.\frac{\partial F}{\partial \kappa}\right|_\beta = \beta\frac{\partial F}{\partial \alpha}. \tag{A49}$$

Similarly, the partial derivatives with respect to independent parameters of λ, μ, and ρ can be derived as:

$$\frac{\partial F}{\partial \lambda} = \frac{1}{2\rho\alpha}\frac{\partial F}{\partial \alpha}, \tag{A50}$$

$$\frac{\partial F}{\partial \mu} = \frac{1}{\rho\alpha}\frac{\partial F}{\partial \alpha} + \frac{1}{2\rho\beta}\frac{\partial F}{\partial \beta}, \tag{A51}$$

$$\left.\frac{\partial F}{\partial \rho}\right|_{\lambda,\mu} = -\frac{\alpha}{2\rho}\frac{\partial F}{\partial \alpha} - \frac{\beta}{2\rho}\frac{\partial F}{\partial \beta} + \frac{\partial F}{\partial \rho}. \tag{A52}$$

Similar results were also given in Tromp *et al.* (2005) where these partial derivatives were called Fréchet kernels in 3-D cases.